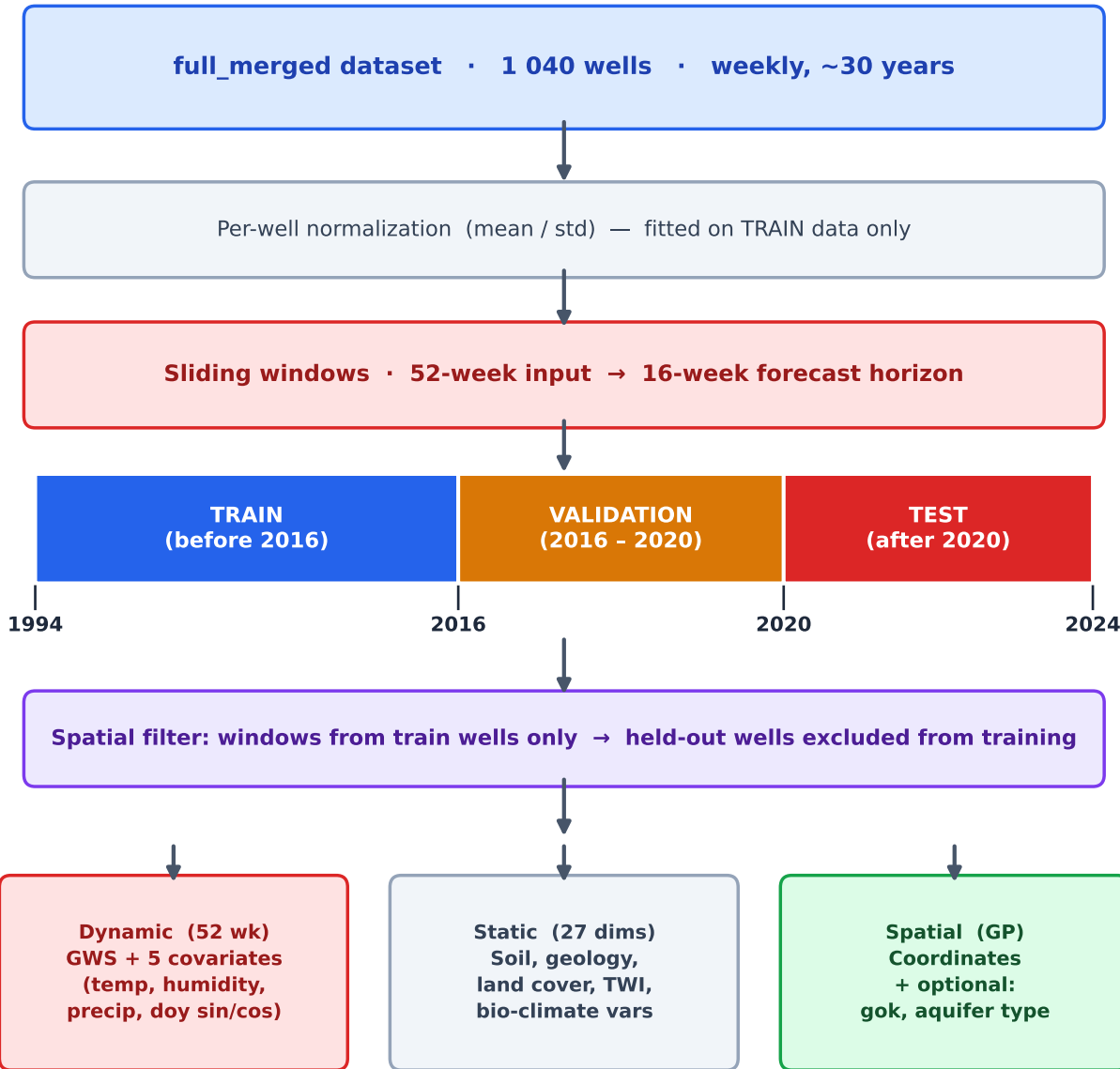


Dataset Structure & Spatial Splits

Temporal structure



Split	Method	Train	Test	Used for
spf0p8	K-means 20 clusters	841 (81%)	199 (19%)	Feature ablation hyperparameter search
spf0p9	K-means 20 clusters	945 (91%)	95 (9%)	Alternative cluster split
random_90	Purely random	936 (90%)	104 (10%)	Random baseline
rmd90	Max-distance ≥ P90	936 (90%)	104 (10%)	Joint GRU+GP evaluation
rmd90_test52	rmd90 merged 95/5	988 (95%)	52 (5%)	Decoupled GRU→GP (fair comparison)
rmd5	Densest clusters	988 (95%)	52 (5%)	Most challenging spatial case

Joint pipeline uses a 3-way split: val wells drawn from train set (val_from_train_fraction = 5%) — test set remains fully held out